

GA.2. - Temă seminar 6

1. $V_1 = \{(x, y, 0) / x, y \in \mathbb{R}\}$

$V_2 = \{(u, 0, v) / u, v \in \mathbb{R}\}$

a) fie $w_1, w_2 \in V_2 \rightarrow u_1 = (u_1, 0, v_1), u_1, v_1 \in \mathbb{R}$
 $x_1, x_2 \in \mathbb{R} \quad w_2 = (u_2, 0, v_2), u_2, v_2 \in \mathbb{R}$

$x_1 w_1 + x_2 w_2 = (x_1 u_1 + x_2 u_2, 0, x_1 v_1 + x_2 v_2)$

$x_1 u_1 + x_2 u_2 \in \mathbb{R}$

$x_1 v_1 + x_2 v_2 \in \mathbb{R}$

$\Rightarrow x_1 w_1 + x_2 w_2 \in V_2 \Rightarrow V_2 \subset \mathbb{R}^3$
 spațiu vectorial

$V_2 \ni (u, 0, v) = u e_1 + v e_3 \Rightarrow B_2 = \{e_1, e_3\}$
 bază
 $u, v \in \mathbb{R}$

$\Rightarrow \dim V_2 = 2$ (nu ~~vectorial~~ vectorial)

b) + dimensiuni (Grassmann)

$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim V_1 \cap V_2$

$V_1 \cap V_2 \ni v \Rightarrow \begin{cases} x = u (= \text{not } t) \\ y = 0 \\ v = 0 \end{cases} \Rightarrow V_1 \cap V_2 = \{(t, 0, 0) / t \in \mathbb{R}\}$
 $= \{t e_1 / t \in \mathbb{R}\} = \langle e_1 \rangle$
 $\Rightarrow \dim V_1 \cap V_2 = 1$

$\dim V_1 = \dim V_2 = 2$

$\Rightarrow \dim(V_1 + V_2) = 2 + 2 - 1 = 3$
 $V_1 + V_2 \subseteq \mathbb{R}^3$
 spațiu vectorial $\Rightarrow V_1 + V_2 = \mathbb{R}^3$

$V_1 \oplus V_2 = \mathbb{R}^3$ nu este directă, deoarece $V_1 \cap V_2 = \langle e_1 \rangle \neq \{0\}$

$$2. \quad V_1 = \{ A \in M_n(\mathbb{R}) \mid \text{Tr } A = 0 \}$$

$$V_2 = \{ A \in M_n(\mathbb{R}) \mid A = x \bar{I}_n, x \in \mathbb{R} \}$$

$$a) \quad \text{fie } A_1, A_2 \in V_1 \Rightarrow \text{Tr } A_1 = 0$$

$$x_1, x_2 \in \mathbb{R} \quad \text{Tr } A_2 = 0$$

$$\text{Tr}(x_1 A_1 + x_2 A_2) = \text{Tr}(x_1 A_1) + \text{Tr}(x_2 A_2) =$$

$$= x_1 \text{Tr } A_1 + x_2 \text{Tr } A_2 = 0 + 0 = 0 \Rightarrow x_1 A_1 + x_2 A_2 \in V_1$$

deci $V_1 \subset M_n(\mathbb{R})$
sub vect

$$\text{fie } A_3, A_4 \in V_2 \Rightarrow A_3 = \lambda_1 \bar{I}_n, \lambda_1 \in \mathbb{R}$$

$$x_3, x_4 \in \mathbb{R} \quad A_4 = \lambda_2 \bar{I}_n, \lambda_2 \in \mathbb{R}$$

$$x_3 A_3 + x_4 A_4 = x_3 \lambda_1 \bar{I}_n + x_4 \lambda_2 \bar{I}_n =$$

$$= (x_3 \lambda_1 + x_4 \lambda_2) \bar{I}_n, x_3 \lambda_1 + x_4 \lambda_2 \in \mathbb{R} \Rightarrow x_3 A_3 + x_4 A_4 \in V_2$$

deci $V_2 \subset M_n(\mathbb{R})$
sub vect

$$b) \quad V_1 + V_2 \subset M_n(\mathbb{R}), \text{ deoarece } M_n(\mathbb{R}) \subset V_1 + V_2$$

$$(b) \quad C \in M_n(\mathbb{R}), (3) \quad A \in V_1, B \in V_2 \text{ a } C = A + B$$

$$\text{notăm } \alpha = \frac{\text{Tr } C}{n} \in \mathbb{R}$$

$$A = C - \alpha \bar{I}_n$$

$$\text{evident: } \text{Tr } A = \text{Tr } C - \text{Tr}(\alpha \bar{I}_n) = \text{Tr } C - \frac{\text{Tr } C}{n} \cdot n = 0$$

$$B = \alpha \bar{I}_n$$

$$\text{luăm: } A = C - \alpha \bar{I}_n$$

$$B = \alpha \bar{I}_n$$

deci (1) $C \in M_n(K)$, (3) $A \in V_1$, $B \in V_2$ a $\bar{C} = A + B \Rightarrow$

$$M_n(K) \subset V_1 + V_2 \Rightarrow M_n(K) = V_1 + V_2$$

arătăm că $V_1 \cap V_2 = \{0_n\}$

$$\text{fie } B \in V_1 \cap V_2 \Rightarrow B = x \bar{I}_n \mid \Rightarrow x=0 \Rightarrow B=0_n$$

$$\text{deci } V_1 \oplus V_2 = M_n(K)$$

c) verif. Th dimensiunii:

$$\dim_K M_n(K) = \dim_K V_1 + \dim_K V_2$$

$$\dim_K M_n(K) = n^2$$

fie $A \in V_1$, $A = (a_{ij})_{i,j=1,\dots,n}$, $\tau_n A = 0$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} = \sum_{1 \leq i,j \leq n} a_{ij} E^{ij}_{ij}, \text{ unde:}$$

$$E^{ij}_{ij} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \text{ pt } i=j \neq n$$

1 pe poz (i, j)
-1 pe poz (n, n)
0 pe celelalte poz

$$\begin{pmatrix} 0 & \dots & 1 & \dots & 0 \\ 0 & 0 & & & 0 \\ 0 & & 0 & & 0 \end{pmatrix}, \text{ pt } i \neq j$$

1 pe poz (i, j)
0 pe celelalte poz

$$\begin{pmatrix} 0 & \dots & 0 \\ \vdots & & \vdots \\ 0 & \dots & 0 \end{pmatrix} = 0_n, \text{ pt } i,j=n$$

0 pe toate poz

$$\mathcal{B}_{V_1} = \{ E^{ij} \}_{1 \leq i, j \leq n} \subset V_1$$

basis

$$\dim V_1 = \text{card } \mathcal{B}_{V_1} = n^2 - 1$$

for $m \in V_2$, $m = (b_{ij})_{1 \leq i, j \leq n}$, $m = \lambda I_n$, $\lambda \in \mathbb{K}$

$$m = \begin{pmatrix} \lambda & 0 & \dots & 0 \\ 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda \end{pmatrix} = \lambda E^{ii}, \text{ unde } E^{ii} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

$$\mathcal{B}_{V_2} = \{ E^{ii} \}_{1 \leq i \leq n} \subset V_2$$

basis

$$\dim V_2 = \text{card } \mathcal{B}_{V_2} = 1$$

anem: $n^2 - 1 + 1 = n^2$ (verif ik dimen)

3. $\mathcal{A}: \mathbb{K}^3 \rightarrow \mathbb{K}^3$

$$\mathcal{A}(x, y, z) = (x + y + z, x - y + z, x - y - z)$$

a) $\mathcal{A}(x) = A x$, unde $x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $\text{ unde } A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix} \in M_3(\mathbb{K})$

for $x_1, x_2, x_3 \in \mathbb{K}^3$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \quad \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$$

unde $x_1, x_2, x_3 \in V_1$

$$\begin{aligned} \mathcal{A}(x_1 x_1 + x_2 x_2 + x_3 x_3) &= A (x_1 x_1 + x_2 x_2 + x_3 x_3) = \\ &= A(x_1 x_1) + A(x_2 x_2) + A(x_3 x_3) = (A x_1) x_1 + (A x_2) x_2 + (A x_3) x_3 = \end{aligned}$$

$$= (\alpha_1 A) x_1 + (\alpha_2 A) x_2 + (\alpha_3 A) x_3 = \alpha_1 (A x_1) + \alpha_2 (A x_2) + \alpha_3 (A x_3) =$$

$$= \alpha_1 \cancel{f}(x_1) + \alpha_2 \cancel{f}(x_2) + \alpha_3 \cancel{f}(x_3) \quad \rightarrow \quad \cancel{f} \text{ not lin}$$

b) $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}$ matrice $\cancel{f}$ not lin $\cancel{f}$ in rep in

base canonica $\dim \mathbb{R}^3$, resp $\mathbb{R}^3$

$\underbrace{\quad}_u \quad \underbrace{\quad}_u \quad \underbrace{\quad}_u$
 $\cancel{f}(e_1) \quad \cancel{f}(e_2) \quad \cancel{f}(e_3)$, unde $\{e_1, e_2, e_3\} \subset \mathbb{R}^3$
 b canonica

$$\cancel{f}(e_1) = \cancel{f}(1, 0, 0) = (1, 1, 1)$$

$$\cancel{f}(e_2) = \cancel{f}(0, 1, 0) = (1, -1, -1)$$

$$\cancel{f}(e_3) = \cancel{f}(0, 0, 1) = (1, 1, -1)$$